TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108
Accredited with NAAC A+ Grade
Approved by AICTE, New Delhi, Govt. of Maharashtra

(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)
Department of Information Technology
Session 2023-24(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

END SEMESTER

Semester: V

Programme: B. Tech Information Technology

Course: Digital Logic & Fundamentals of Microprocessor

Time: 3 Hours **Max Marks:** 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BEC33510	Course Outcome
CO1	To classify the working principles of various combinational circuits and their applications in different digital systems.
CO2	To analyze and recognize various sequential components utilized in the design of combinational circuits.
CO3	To develop the skills to design and validate the operation of various combinational and sequential circuits.
CO4	o provide knowledge on the 8085 microprocessor, covering its architecture, functionalities, addressing modes, port types, and practical applications.
CO5	To explore the use of Assembly language programming for the 8085 microprocessor, emphasizing its timing, control features in industrial embedded systems.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	State De Morgan’s Theorem	CO1	2	2
b.	Define Multiplexer.	CO2	2	2
c.	What is Flip-Flop?	CO3	2	2
d.	Define addressing modes in 8085.	CO4	2	2

e.	What is interrupt?	CO5	3	2
	Group B			
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Simplify using Boolean algebra: $A+A'B+AB'$	CO1	2	5
b.	Explain SOP and POS forms with examples.	CO1	3	5
c.	Minimize given function using K-map (4 variable).	C01	2	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain working of Multiplexer with diagram.	CO2	2	5
b.	Design a 4-bit binary adder using logic gates.	CO2	4	5
c.	Compare Encoder and Decoder with examples.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain SR flip-flop with truth table.	CO3	2	5
b.	Describe working of JK flip-flop and its race condition.	CO3	3	5
c.	Explain shift registers and their types.	CO3	4	5
Que. 5	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Draw and explain 8085 architecture.	CO4	3	5
b.	Explain different addressing modes with examples.	CO4	3	5
c.	Write note on instruction set classification.	C04	4	5
Que. 6	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain types of interrupts in 8085.	CO5	3	5
b.	Draw timing diagram of memory read operation.	CO5	4	5
c.	Explain basic memory organization of 8085.	C05	2	5

Bloom’s Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create